
Relations Need a Home: A Relational Binding Interface for Spatial Reasoning in Frozen Vision–Language Models

Haziq Mohammad Khalid
@100601

Hamad Siddiqi
@101467

Computer Vision (COE486) Project

Abstract

Vision-language models (VLMs) reliably report *where* an object is, yet falter when asked how two objects relate: whether one is left of another, above it, or in front of it. We trace this gap to relational binding, a failure distinct from perception and from reasoning in general. A spatial relation is a property of a *pair* of entities, and a decoder-only transformer has no representational slot in which such a pair-property can live. Two findings support this. Reading a coordinate directly from the model’s own visual attention answers spatial-relation questions at 84.5% accuracy versus the model’s own 64.8%, so the positions it needs are already present in its activations. Yet feeding those positions back into the individual object tokens does not build the relation and improve performance significantly. We find that stating the relation symbolically, improves performance because the symbol gives the answer a token to attend to. So we give the relation a home: a single relational-cross-attention block whose values are content-independent learned symbols, in the sense of the relational bottleneck. It reads the coordinates the model already computes and emits one *relation token* that we insert into the residual stream, where the frozen backbone’s pretrained reasoning consumes it. The backbone is never fine-tuned; only the interface is trained ($\approx 0.26\%$ the size of the backbone) by distilling the symbolic intervention. On a CV-Bench spatial-relation task the interface improves performance by +9.9 points ($67.2\% \rightarrow 77.1\%$).

1 Introduction

Vision-language models have improved quickly on captioning, visual question answering, and open-ended visual dialogue, however they continually struggle to reason about the spatial relations between objects Tong et al. [2024a], Kamath et al. [2023], Liu et al. [2023], Fu et al. [2024], Tong et al. [2024b]. A model that names every object in a scene, and can point to each, still answers near chance when asked whether one object lies to the left of another, or above it. We argue that the cause of this failure has been mislocated.

There are two current explanations for this weakness in VLMs. The first blames perception: AdaptVis Chen et al. [2025b] attributes the errors to misallocated visual attention and reweights the attention distribution at inference. The second blames downstream computation: deep layers attend to the correct regions even when the answer is wrong Liu et al. [2025], which locates the fault in later computation rather than in where the model looks. A third group recovers spatial behavior without retraining, by amplifying or normalizing the spatial signal in token embeddings Qi et al. [2025], Cui

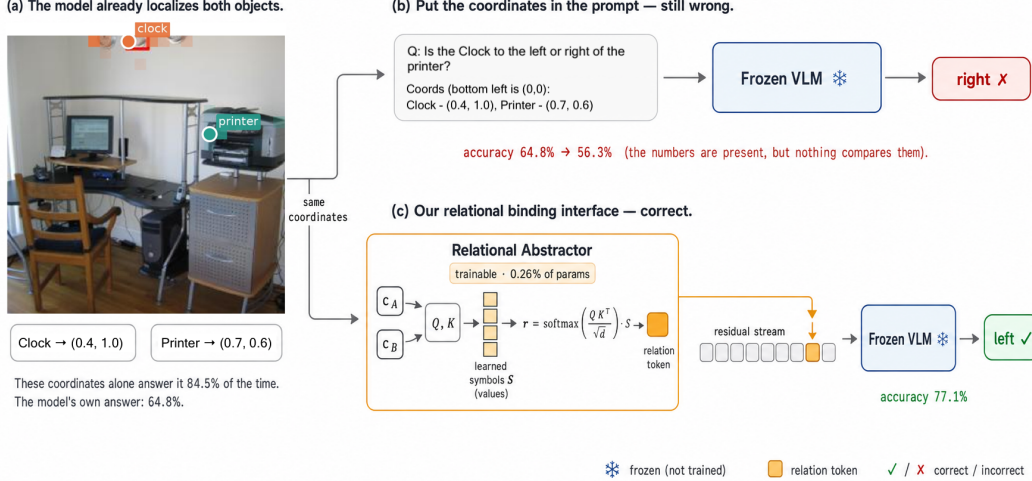


Figure 1: (a) The frozen VLM’s own attention already localizes both objects, and those coordinates answer the relation question 84.5% of the time versus the model’s own 64.8%, so perception is not the bottleneck. (b) Handing the model the very same coordinates as text in the prompt does not help and in fact hurts (64.8% → 56.3%): the numbers are present, but the decoder has no operation that compares two coordinate tokens. (c) Our relational binding interface reads the same coordinates, forms a single *relation token* through a relational bottleneck, and inserts it into the residual stream, where the frozen backbone reads it and answers correctly (77.1%). Only the interface is trained ($\approx 0.26\%$ the size of the backbone).

et al. [2026] or re-indexing the positional code of visual tokens Ge et al. [2024], Li et al. [2025]. All these methods help by a few points, and they share one property: each method one acts on a single token. It changes where the token attends, scales a direction inside it, or alters how its position is encoded.

We reach a different diagnosis, and two experiments motivate it. First, we read a coordinate from the model’s own noun-token attention over the image and answer the spatial question from that coordinate alone. Accuracy rises from 64.8% to 84.5%, with both objects localized on every item, so the positions the task needs are present in the model’s activations. Perception is not the bottleneck. Second, we feed those coordinates back in through the object tokens, and the model gains no relation. Raw coordinate tokens *lower* accuracy. An absolute position vector per object token changes nothing significant. And a relative contrastive scheme that pushes the two tokens apart along their true displacement gives a small bump but we show that to be due to non-relational factors. A relation such as “A is left of B” is a property of the pair (A, B) : biasing two object tokens leaves two per-object signals the model must still compare, and that comparison is the step it cannot take unaided. The one intervention that works, stating the relation symbolically (for example, inserting “A is to the left of B”), lifts accuracy sharply, because the inserted symbol becomes a token the answer can attend to. What the model lacks is a place to put the relation: a single position in the sequence whose content is the relation itself, a binding site.

This points to a method. Rather than write the relation as a fragile natural-language string, we compute it with a dedicated module and give it a home in the residual stream. The module is one relational-cross-attention block Altabaa et al. [2024]: queries and keys are projected from object encodings and their inner products form a relation matrix, while the values are content-independent learned symbols, the defining move of the relational bottleneck Webb et al. [2024], which forces the output to reflect only how the inputs relate. We build the encodings from the coordinates the model already computes, collapse the block to a single relation token, and add it at a reserved position in the stream, where the frozen backbone reads it through ordinary attention. The backbone is never fine-tuned; we train only the interface ($\approx 0.26\%$ the size of the backbone) by distilling the symbolic intervention above. Frozen language models already carry emergent symbolic machinery, including symbol-abstraction, induction, and retrieval heads over symbol-like representations Yang et al. [2025a].

Our evaluation separates genuine relational gains from answer leakage. The headline task is the CV-Bench spatial-relation judgment (left/right/above/below), whose answer follows from image-plane coordinates but is reached only through a pairwise comparison the model cannot make unaided, so a gain there is relational, not tautological. A position-swap control exchanges the two coordinates and flips the relation, which should invert any honest gain. Two 3D tasks, Depth (“closer to the camera”) and Distance (“closer to a reference”), act as negative controls: neither can be solved from image-plane position, so a coordinate-driven token should do nothing on them in ideal cases.

Contributions.

- We relocate the cause of VLM spatial failure from perception or reasoning to relational binding, we prove this by showing that attention coordinates already attend to the correct parts of the image (+19.7 accuracy improvement if answer is just derived from the attention coordinates) and a negative prompt injection result (feeding these coordinates into the prompt is detrimental to performance).
- We introduce a frozen-backbone relational binding interface: a relational-bottleneck module that reads the model’s own coordinates and writes a single, content-isolated relation token into the residual stream, and trains $\approx 0.26\%$ of the backbone’s parameters with no fine-tuning.
- We design a protocol (a swap-sensitive headline, a position-swap control, and two 3D tasks unsolvable from the image plane) that separates relational gains from answer leakage, and show a relational bottleneck can act as an interface that grounds in and returns to a frozen foundation model, rather than the standalone network it has been in prior work.

2 Related Work

Diagnosing spatial failure in VLMs. AdaptVis Chen et al. [2025b] ties the errors to misdirected visual attention and corrects the attention distribution by confidence. Another blames downstream computation: deep layers attend to the right regions even when the answer is wrong Liu et al. [2025]. Benchmarks including CV-Bench Tong et al. [2024a], What’s Up Kamath et al. [2023], VSR Liu et al. [2023], BLINK Fu et al. [2024], and MMVP Tong et al. [2024b] have made the deficit reproducible. Our paper argues that perception is intact, since the needed coordinates are recoverable from the model, and reasoning is available once the right input arrives. What is absent is a representational slot for a pair-property. Where other methods would reweight existing attention or computation, we build a new representation.

Inference-time interventions in embedding and positional space. “Beyond Semantics” Qi et al. [2025] locates spatial content in embedding norms and mid layers and exploits it through normalization and mid-layer extraction; “Dual Mechanisms” Cui et al. [2026] finds spatial information spread across tokens and amplifies the vision-derived part; V2PE Ge et al. [2024] and RoPE re-scaling Li et al. [2025] change how position is encoded for the image stream. Each acts on a single token’s features or on its positional foundation, and none can produce a relation, which is a function of two tokens, not a per-token quantity to amplify, normalize, or re-encode. Our interface takes position as input and returns a relation as output. Its content-independent values guarantee the output encodes how the tokens relate instead of reshaping a copy of either.

Added tokens, visual prompts, and backbone fine-tuning. ViP-LLaVA Cai et al. [2024] blends visual prompts onto pixels and instruction-tunes; LLaVA-SP Lou et al. [2025] appends pooled “visual spatial tokens” and fine-tunes with LoRA; Set-of-Mark Yang et al. [2023] overlays numbered marks; CLOC Chen et al. [2025a] adds promptable localization embeddings during pretraining. These differ from ours in three ways: location (pixel or early visual stream vs. the LLM residual stream), backbone (fine-tuned or re-pretrained vs. frozen), and payload (object content such as region features or marks vs. a content-isolated relation). The last matters most: adding content-bearing tokens still leaves the cross-object comparison to the backbone, which is the step we identify as missing.

Relational bottlenecks, Abstractors, and symbolic mechanisms. Our module instantiates the relational-bottleneck principle Webb et al. [2024], realized before in ESNB Webb et al. [2021], CoRelNet Kerg et al. [2022], and the Abstractor’s relational cross-attention Altabaa et al. [2024], with hyperdimensional variants Mejri et al. [2024a,b]. We borrow the mechanism unchanged; the novelty

is in deployment. In prior work the bottleneck *is* the network, trained from scratch on synthetic tasks. Here it is used as an interface to a frozen multimodal model: its inputs are coordinates the host perceives, and its output returns to the host’s residual stream so the host’s own reasoning finishes the task, with no bespoke decoder. The choice follows two findings: frozen LLMs host emergent symbol abstraction, induction, and retrieval heads Yang et al. [2025a], and transformers route sensory but not relational information Altabaa and Lafferty [2025]. We supply the missing relational signal in a form the existing heads can read.

Activation steering and latent reasoning tokens. Contrastive activation steering Rimsky et al. [2024] adds a fixed direction to the residual stream. Our results show this cannot manufacture a genuine relation, since a relation is not already a direction in a single token’s activation. Coconut Hao et al. [2024] feeds the model’s hidden state back as the next embedding and Mirage Yang et al. [2025b] trains latent “mental imagery” tokens, but in both the latent tokens are free-form and come from the backbone. Our relation token comes from outside the backbone with fixed structure and occupies one reserved slot, which is what lets us isolate it, swap it as a control to see if the gain is relational.

3 Methodology

We intervene on a *frozen* LLaVA-1.5-7B Liu et al. [2024] at inference time, never updating its weights. Its CLIP ViT-L/14-336 encoder Radford et al. [2021] yields a 24×24 patch grid in exact one-to-one correspondence with the 576 visual tokens the decoder consumes. This alignment lets us localize objects in the same coordinate frame the decoder reads, and an inference-time intervention can be toggled or ablated on a single forward pass.

Reading object coordinates from the model. Both the diagnostic experiments and the interface need a single (x, y) per referenced object in the decoder’s token grid. The referenced objects come from the CV-Bench question templates: on Relation the two related nouns are parsed from the question, on Depth they are the two answer choices, and on Distance the reference object joins them, for at most three per item. We then take a single forward pass with attentions retained and read, at layer 14, each phrase’s attention over the 576 visual positions, averaged over heads and over the phrase’s own tokens. In order to get a clean visual of object maps rather than general attention habits of LLaVA, we subtract the mean attention-to-image over all text positions, which removes position bias and attention sinks, and we zero the bottom 0.75 quantile of the mass that remains. The centroid of what survives is the object’s coordinate, in $[0, 1]^2$ with the origin at the bottom left. Nothing here is trained and nothing is read from outside the model. These coordinates feed both the oracle (Sec. 4.1) and the interface.

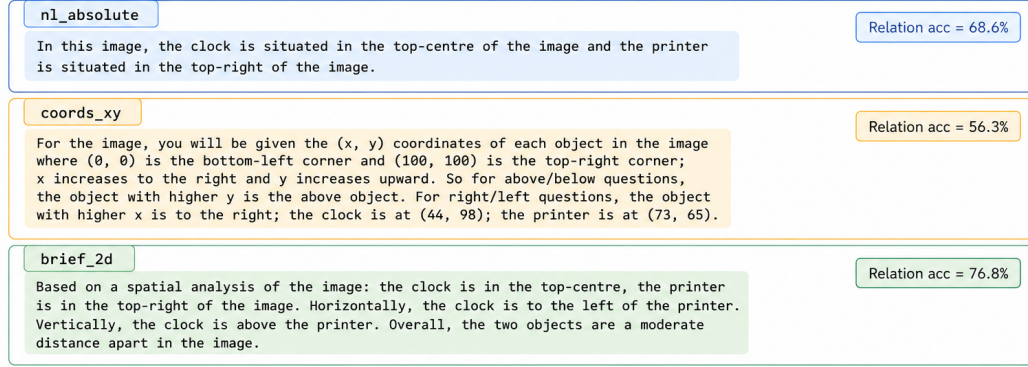
Position-injection diagnostics. To test whether the model can use these coordinates once they are handed back, we deliver the *same* positional information in several forms (example shown in Figure 2): the raw (x, y) written as text tokens (coords_xy), a natural-language statement of each object’s position (nl_absolute), an absolute position vector added to each object token (steer_abs), contrastive position vectors added to each object token pushing the two in opposite directions (steer_rel), and a brief summary of the 2D scene written in natural language (brief_2d).

The relational binding interface. The interface is a single relational-cross-attention block Altabaa et al. [2024]. Each object’s normalized coordinate is mapped to a Fourier positional encoding and combined with a learned role embedding marking first- vs. second-mentioned object; queries and keys are projected from these encodings and their inner products form the relation, while the *values* are a small set of content-independent learned symbols, the relational bottleneck Webb et al. [2024], which forces the output to encode how the objects relate rather than what they are. The block collapses to one 4096-dimensional relation token, written into the residual stream at a reserved slot at layer 14 and read through ordinary attention. The module is 17.9M parameters ($\approx 0.26\%$ of the backbone). Appendix B gives the block, the write rule, and what they inherit from and change about prior uses of the bottleneck.

Training. Every two-choice example is paired with a counterfactual twin whose coordinates are exchanged and label flipped, which no swap-invariant module can fit. This augmentation allows us to

CV-Bench Relation item

Prompt-space interventions — prepend text, then the question



Activation-space interventions — prompt unchanged; edit the residual stream at layer 14

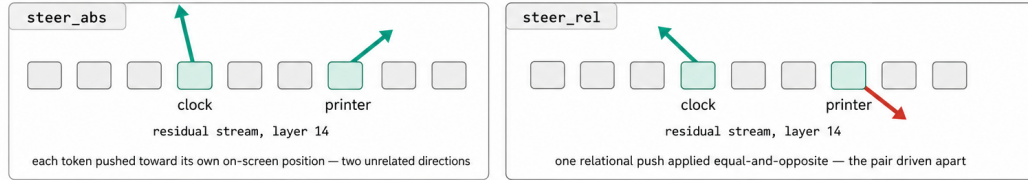


Figure 2: Example prompt for the five different types of ways we evaluate giving the model its attention read coordinates. Prompt remains unchanged for Activation-space interventions.

double our training dataset size. Because a coordinate-driven relation token is only appropriate when the answer is an image-plane relation, a router decides per query whether to inject it. All generation is greedy for reproducibility. Appendix B.3 shows the training curve and gives the objective and the hyperparameters.

4 Results

Setup. The backbone is a frozen LLaVA-1.5-7B; no backbone weight is updated. Spatial questions come from CV-Bench Tong et al. [2024a]. The headline task is the two-object spatial *relation* judgment (left/right/above/below), with a fixed split of 454 train, 65 validation, and 131 test items. The interface is fit on train, selected on validation, and reported on the untouched test split. Depth ($n=300$) and Distance ($n=300$) are 3D negative controls, since their answers do not follow from image-plane position. We report accuracy with 95% bootstrap confidence intervals (10,000 resamples), the paired change Δ against the frozen model with its CI, and, for coordinate-reading interventions, a position-swap control that exchanges the two coordinates so the true relation flips. The swap-sensitive quantity *Real-Swap* is the most direct test of whether an effect is relational.

4.1 The coordinates the model needs are already there

We first ask whether perception is the bottleneck. From the frozen model’s own noun-token attention we read one (x, y) per object and answer the relation directly from those two coordinates. The readout is far more accurate than the model itself (Table 1): it reaches 84.5% against the model’s 64.8%, a +19.7-point gap, and on swapped coordinates falls to 15.5%, so it reads the relation rather than a prior. The positions the task requires are present and correct in the model’s activations, the model just fails to convert them into a relation.

4.2 Injecting positions into tokens does not build a relation

Can we make the model use these coordinates by putting them back into the object tokens? We test four per-token injections and one symbolic statement on the same 650 relation items (Table 2, Fig. 3).

	Accuracy
Frozen model (vanilla)	64.8% [61.1, 68.5]
Coordinate oracle	84.5%
Coordinate oracle, coordinates swapped	15.5%

Table 1: **Localization oracle on the relation task** ($n=650$). Answering directly from the two coordinates read out of the frozen model’s own attention far exceeds the model’s accuracy and inverts under a coordinate swap. Perception is not the bottleneck.

Raw coordinate tokens (COORDS_XY) are actively harmful, -8.5 points ($p=10^{-4}$): the numbers are present but the model has no operation that turns two coordinate tokens into a comparison. Stating each position in words (NL_ABSOLUTE) helps marginally ($+3.8$), and an absolute position vector per token (STEER_ABS) is indistinguishable from nothing ($+2.2$). Relative contrastive steering (STEER_REL) raises raw accuracy by 3.7 points, however the swap control leaves accuracy at 56.8% , a *Real-Swap* of only $+11.7$. Therefore relative contrastive steering does not track relation strongly, it is just a generic nudge due to the fact that it survives having the relation reversed. Only the symbolic statement behaves like a relation. A brief clause naming it (BRIEF_2D) lifts accuracy by 12.0 points to 76.8% ($p < 10^{-9}$), and the swap does not merely erase but inverts the gain: accuracy drops to 21.8% , a *Real-Swap* of $+54.9$. The contrast between STEER_REL (raw gain, no swap sensitivity) and BRIEF_2D (large, swap-sensitive gain) is what drives our motivation, a relation needs a token of its own, and steering the object tokens cannot supply one.

Method	Acc.	Δ vs. vanilla (95% CI)	Swap acc.	Real-Swap
vanilla	0.648	—	—	—
nl_absolute	0.686	$+0.038$ [$+0.006, +0.072$]	—	—
coords_xy	0.563	-0.085 [$-0.129, -0.042$]	0.546	$+0.017$
steer_abs	0.669	$+0.022$ [$-0.014, +0.057$]	—	—
steer_rel	0.685	$+0.037$ [$+0.006, +0.068$]	0.568	$+0.117$
brief_2d	0.768	$+0.120$ [$+0.083, +0.158$]	0.218	$+0.549$

Table 2: **Position injections vs. a symbolic statement on the relation task** ($n=650$). Δ is the paired change against the frozen model with a 95% CI. *Swap acc.* and *Real-Swap* are reported for coordinate-reading conditions. Only stating the relation (BRIEF_2D) produces a large, swap-sensitive gain.

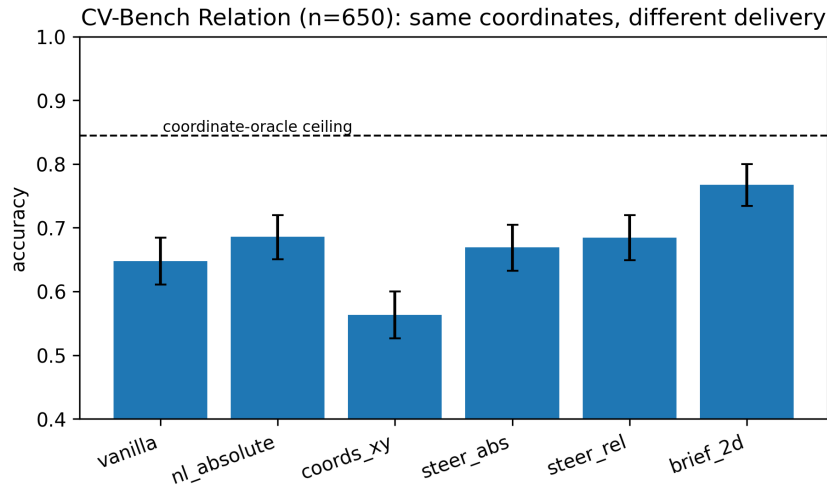


Figure 3: **Relation task, real vs. swapped.** Bars are accuracy against the frozen baseline (dashed); markers show accuracy after the coordinate swap. Interventions that encode the relation (BRIEF_2D, the learned ABTRACTOR token) rise on the real inputs and fall below chance on the swap; per-token steering does not separate the two.

4.3 A learned relation token: the relational binding interface

Our diagnosis says the relation has to arrive as a token, so we build one and hand it over at a single reserved slot. On the held-out test split (Table 3) the frozen model scores 67.2%; the relation token raises it to 77.1%, +9.9 points ($p \approx 0.04$), and exchanging the two coordinates drops it to 29.8%, a *Real-Swap* of +47.3. The token reproduces the swap signature of the symbolic teacher it distills (72.5%, +5.3 but not significant on 131 items) while reaching significance from one fixed slot rather than an inserted phrase. It has learned to write the relation, not the objects.

That signature is built into the objective rather than found: each training item is paired with a twin whose coordinates are exchanged and label flipped, so nothing swap-invariant fits the data at all. The token is also distilled from BRIEF_2D rather than fit to the gold label, which is what makes it train through a frozen backbone. Both choices are in Appendix B.3.

Table 3: **The relational binding interface on the held-out test split** ($n=131$). The learned relation token (ABTRACTOR) improves the frozen model significantly and collapses below chance under the swap control, and matches the swap signature of the symbolic teacher it distills.

Method	Acc.	Δ vs. vanilla (95% CI)	Swap acc.	Real-Swap
vanilla (frozen)	0.672	—	—	—
brief_2d (teacher)	0.725	+0.053 [−0.031, +0.137]	0.252	+0.473
abstractor (interface)	0.771	+0.099 [+0.008, +0.191]	0.298	+0.473

4.4 Specificity and routed deployment

A relation token grounded in image-plane coordinates should help only when the answer is an image-plane relation, and the cross-task deltas confirm it is not a generic booster (Fig. 4). The symbolic statement that gives +12.0 on relation gives −22.0 on Depth and −2.0 on Distance; raw coordinate tokens cost 24.0 on Depth. These 3D tasks cannot be read off the image plane (Distance sits at chance, 51.0%), so a coordinate-driven signal can only mislead them. The honest interventions help the relation task and harm the controls, the pattern expected of a relational effect but not a generic boost.

Because a coordinate-driven token is harmful when misapplied, deployment needs a gate, and the gate is trivial: inject only when every answer choice is an image-plane direction word. On CV-Bench the rule is exact rather than approximate. Relation’s choices are always direction words, and Depth’s and Distance’s are always object names, so the router fires on every relation item and on no control item by construction, a gated intervention keeps its gain on the image plane at exactly zero cost on the tasks it cannot answer, where always injecting would cost 22.0 points on Depth. The interface is a net gain across the suite only because it knows when to keep quiet. What the gate cannot do is extend the method, it protects Depth and Distance by declining them, which leaves the 3D relations to Sec. 5.

5 Future Work

One interface, several relations. The module is fixed to one relation family, and it can afford to be, because with only one family the relation being asked for is never in doubt. Hosting several requires the module to be told which one, and the bottleneck constrains how. Object content cannot enter the values without dissolving the constraint, but the relation type is not object content — it is a property of the question — so it can select among symbols without loosening anything: a bank per family with the query choosing, or a type embedding added to the role embedding. The evaluation carries over unchanged. The swap control is defined for any asymmetric relation, and a family whose token survives having its relation reversed is a family the token is not representing.

More datasets. An important caveat of our results is that CV-Bench relation questions often come with objects boxed in a red square. Some CV-Bench questions require the models to distinguish between multiple instances of the same object in the image via the red square. We do not account for this in our method, if multiple instances of the same object are in the image then we rely on the attention maps + centroid selection to come up with a general average location of the objects. This

Where each delivery helps — and where it does damage

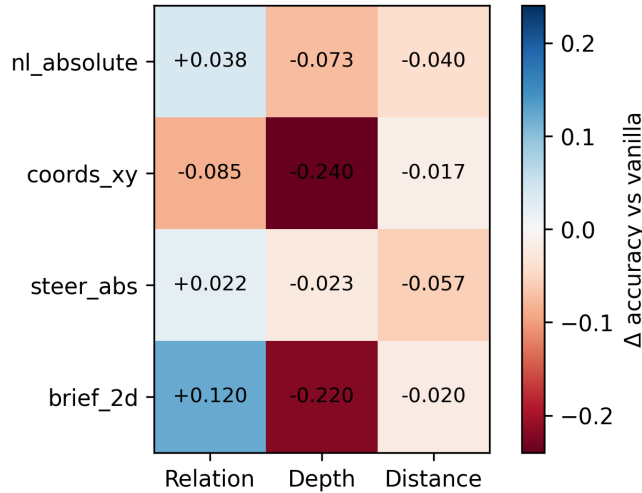


Figure 4: **Cross-task specificity.** Change vs. the frozen model on Relation, Depth, and Distance. Coordinate-driven interventions help the image-plane relation task and harm the 3D controls, so the effect is relational, not a general boost.

clearly is a large limitation and source of error in the CV-bench dataset thus extending the method to other datasets where only one instance of each object exists in the image could lead to better results.

3D relations from 2D evidence. The token fails Depth and Distance because a centroid cannot express them however a centroid is not all the model gives us. The extent of an object’s attention blob is a size cue, its lowest attended row is a ground-contact cue, and the overlap of two blobs is an occlusion cue, all three can be seen as 3D evidence computable from the image plane. The interface would not change. Only the encoding widens, from a coordinate to the handful of 2D observables the frozen host already produces. This would allow us to generalize our method to broad use cases.

6 Conclusion

We located why VLMs that can point to every object still fail to say how two objects relate. The failure is one of *relational binding*, the model perceives both objects and could reason about a relation once it is stated, but has no representational slot in which to form the relation itself. The coordinates it needs are already in its attention; reading them out answers the question at 84.5% against 64.8%. And once the relation is handed over as a token the model can attend to, its existing reasoning finishes the job. What the decoder-only architecture lacks is a place to put a property of a *pair* of tokens, and our swap control separates a generic nudge from a genuine relational signal that per-token steering cannot produce.

Acting on that diagnosis, we built a relational binding interface: one relational-cross-attention block with content-independent learned symbols, at $\approx 0.26\%$ of the backbone and with the backbone frozen. It raises LLaVA-1.5-7B by 9.9 points on a held-out relation split, and the gain is relational rather than cosmetic: it inverts under a coordinate swap, is confined to the image-plane task, and, when routed, leaves the 3D controls untouched.

The broader claim is architectural. A relational bottleneck has been, in prior work, an entire network trained end to end on synthetic tasks; here it is an *interface* that grounds in a frozen model’s own perception and returns to its own reasoning, with no external decoder and no weight update.

References

Awni Altabaa and John Lafferty. Disentangling and integrating relational and sensory information in transformer architectures. In *Proceedings of the 42nd International Conference on Machine*

- Learning (ICML)*, volume 267 of *Proceedings of Machine Learning Research*, pages 1271–1297, 2025. URL <https://arxiv.org/abs/2405.16727>.
- Awni Altabaa, Taylor Webb, Jonathan Cohen, and John Lafferty. Abstractors and relational cross-attention: An inductive bias for explicit relational reasoning in Transformers. In *The Twelfth International Conference on Learning Representations (ICLR)*, 2024. URL <https://arxiv.org/abs/2304.00195>.
- Mu Cai, Haotian Liu, Siva Karthik Mustikovela, Gregory P. Meyer, Yuning Chai, Dennis Park, and Yong Jae Lee. ViP-LLaVA: Making large multimodal models understand arbitrary visual prompts. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 12914–12923, 2024.
- Hong-You Chen, Zhengfeng Lai, Haotian Zhang, Xinze Wang, Marcin Eichner, Keen You, Meng Cao, Bowen Zhang, Yinfei Yang, and Zhe Gan. Contrastive localized language-image pre-training. In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, volume 267 of *Proceedings of Machine Learning Research*, 2025a. URL <https://arxiv.org/abs/2410.02746>.
- Shiqi Chen, Tongyao Zhu, Ruochen Zhou, Jinghan Zhang, Siyang Gao, Juan Carlos Niebles, Mor Geva, Junxian He, Jiajun Wu, and Manling Li. Why is spatial reasoning hard for VLMs? An attention mechanism perspective on focus areas. In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, volume 267 of *Proceedings of Machine Learning Research*, 2025b. URL <https://arxiv.org/abs/2503.01773>.
- Kelly Cui, Nikhil Prakash, Ayush Raina, David Bau, Antonio Torralba, and Tamar Rott Shaham. The dual mechanisms of spatial reasoning in vision-language models. *arXiv preprint arXiv:2603.22278*, 2026.
- Xingyu Fu, Yushi Hu, Bangzheng Li, Yu Feng, Haoyu Wang, Xudong Lin, Dan Roth, Noah A. Smith, Wei-Chiu Ma, and Ranjay Krishna. BLINK: Multimodal large language models can see but not perceive. In *Proceedings of the European Conference on Computer Vision (ECCV)*, pages 148–166, 2024. doi: 10.1007/978-3-031-73337-6_9.
- Junqi Ge, Ziyi Chen, Jintao Lin, Jinguo Zhu, Xihui Liu, Jifeng Dai, and Xizhou Zhu. V2PE: Improving multimodal long-context capability of vision-language models with variable visual position encoding. *arXiv preprint arXiv:2412.09616*, 2024.
- Shibo Hao, Sainbayar Sukhbaatar, DiJia Su, Xian Li, Zhiting Hu, Jason Weston, and Yuandong Tian. Training large language models to reason in a continuous latent space. *arXiv preprint arXiv:2412.06769*, 2024.
- Amita Kamath, Jack Hessel, and Kai-Wei Chang. What’s “up” with vision-language models? Investigating their struggle with spatial reasoning. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 9161–9175, 2023. doi: 10.18653/v1/2023.emnlp-main.568. URL <https://aclanthology.org/2023.emnlp-main.568>.
- Giancarlo Kerg, Sarthak Mittal, David Rolnick, Yoshua Bengio, Blake Richards, and Guillaume Lajoie. On neural architecture inductive biases for relational tasks. *arXiv preprint arXiv:2206.05056*, 2022.
- Yueyan Li, Chenggong Zhao, Zeyuan Zang, Caixia Yuan, and Xiaojie Wang. Reading images like texts: Sequential image understanding in vision-language models. *arXiv preprint arXiv:2509.19191*, 2025.
- Fangyu Liu, Guy Emerson, and Nigel Collier. Visual spatial reasoning. *Transactions of the Association for Computational Linguistics*, 11:635–651, 2023. doi: 10.1162/tacl_a_00566.
- Haotian Liu, Chunyuan Li, Yuheng Li, and Yong Jae Lee. Improved baselines with visual instruction tuning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 26296–26306, 2024. URL <https://arxiv.org/abs/2310.03744>.

- Zhining Liu, Ziyi Chen, Hui Liu, Chen Luo, Xianfeng Tang, Suhang Wang, Joy Zeng, Zhenwei Dai, Zhan Shi, Tianxin Wei, Benoit Dumoulin, and Hanghang Tong. Seeing but not believing: Probing the disconnect between visual attention and answer correctness in VLMs. *arXiv preprint arXiv:2510.17771*, 2025.
- Haoran Lou, Chunxiao Fan, Ziyang Liu, Yuexin Wu, and Xinliang Wang. LLaVA-SP: Enhancing visual representation with visual spatial tokens for MLLMs. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2025. URL <https://arxiv.org/abs/2507.00505>.
- Mohamed Mejri, Chandramouli Amarnath, and Abhijit Chatterjee. LARS-VSA: A vector symbolic architecture for learning with abstract rules. *arXiv preprint arXiv:2405.14436*, 2024a.
- Mohamed Mejri, Chandramouli Amarnath, and Abhijit Chatterjee. RESOLVE: Relational reasoning with symbolic and object-level features using vector symbolic processing. *arXiv preprint arXiv:2411.08290*, 2024b.
- Jianing Qi, Jiawei Liu, Hao Tang, and Zhigang Zhu. Beyond semantics: Rediscovering spatial awareness in vision-language models. *arXiv preprint arXiv:2503.17349*, 2025.
- Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. Learning transferable visual models from natural language supervision. In *Proceedings of the 38th International Conference on Machine Learning (ICML)*, volume 139 of *Proceedings of Machine Learning Research*, pages 8748–8763, 2021.
- Nina Rimsky, Nick Gabrieli, Julian Schulz, Meg Tong, Evan Hubinger, and Alexander Turner. Steering Llama 2 via contrastive activation addition. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 15504–15522, 2024. doi: 10.18653/v1/2024.acl-long.828.
- Shengbang Tong, Ellis Brown, Penghao Wu, Sanghyun Woo, Manoj Middepogu, Sai Charitha Akula, Jihan Yang, Shusheng Yang, Adithya Iyer, Xichen Pan, Austin Wang, Rob Fergus, Yann LeCun, and Saining Xie. Cambrian-1: A fully open, vision-centric exploration of multimodal LLMs, 2024a. URL <https://arxiv.org/abs/2406.16860>.
- Shengbang Tong, Zhuang Liu, Yuexiang Zhai, Yi Ma, Yann LeCun, and Saining Xie. Eyes wide shut? Exploring the visual shortcomings of multimodal LLMs. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 9568–9578, 2024b.
- Taylor W. Webb, Ishan Sinha, and Jonathan D. Cohen. Emergent symbols through binding in external memory. In *International Conference on Learning Representations (ICLR)*, 2021. URL <https://arxiv.org/abs/2012.14601>.
- Taylor W. Webb, Steven M. Frankland, Awni Altabaa, Simon Segert, Kamesh Krishnamurthy, Declan Campbell, Jacob Russin, Tyler Giallanza, Randall O’Reilly, John Lafferty, and Jonathan D. Cohen. The relational bottleneck as an inductive bias for efficient abstraction. *Trends in Cognitive Sciences*, 28(9):829–843, 2024. doi: 10.1016/j.tics.2024.04.001.
- Jianwei Yang, Hao Zhang, Feng Li, Xueyan Zou, Chunyuan Li, and Jianfeng Gao. Set-of-mark prompting unleashes extraordinary visual grounding in GPT-4V. *arXiv preprint arXiv:2310.11441*, 2023.
- Yukang Yang, Declan Iain Campbell, Kaixuan Huang, Mengdi Wang, Jonathan D. Cohen, and Taylor Whittington Webb. Emergent symbolic mechanisms support abstract reasoning in large language models. In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, volume 267 of *Proceedings of Machine Learning Research*, pages 70515–70549, 2025a. URL <https://arxiv.org/abs/2502.20332>.
- Zeyuan Yang, Xueyang Yu, Delin Chen, Maohao Shen, and Chuang Gan. Machine mental imagery: Empower multimodal reasoning with latent visual tokens. *arXiv preprint arXiv:2506.17218*, 2025b.

A Project Timeline

Weeks 1 – 3: Literature review and small proof of concept experiments. Kaggle notebooks provides 30 hours of free access to GPUs, we had 3 accounts and spent our limit on all for two of those three weeks. That alone sums to 180 hours of small experiments running in order to fine tune our idea. The idea started as a uncertainty quantification method for VLMs however we realized that attention maps actually attended to the correct places and pivoted to our eventual paper.

Rest of the weeks: Implementation, debugging, finetuning and paper/presentation writing.

B The relational binding interface

B.1 What we take from prior work

The module is one block of relational cross-attention (RCA), the mechanism introduced with the Abstractor Altabaa et al. [2024]. Over a set of objects $X = (x_1, \dots, x_N)$, ordinary attention returns a mixture of transformed inputs,

$$E_i = \sum_j \text{softmax}_j \left(\frac{\langle \phi_q(x_i), \phi_k(x_j) \rangle}{\sqrt{d}} \right) \phi_v(x_j),$$

so E_i carries the features of whatever it attended to. RCA keeps the query-key term and replaces the values with learned symbols $S = (s_1, \dots, s_N)$ that do not depend on X :

$$A_i = \sum_j \text{softmax}_j \left(\frac{\langle \phi_q(x_i), \phi_k(x_j) \rangle}{\sqrt{d}} \right) s_j.$$

The inner products are now the only path from X to the output. A_i can say how object i stands to the others, it cannot say what any of them is. This is the relational bottleneck Webb et al. [2024].

We take RCA unchanged. What changes is everything around it. In each of those three the bottleneck *is* the network: trained from scratch on a synthetic task, with a decoder trained beside it that learns to read whatever code the symbols settle into. Ours is an interface. Its inputs are coordinates a frozen host already computes, and its output goes back to that host, which will not adapt to it. The rest of this appendix is the consequences of that one difference.

B.2 The module

Encoding. An item references N objects ($N=2$ on Relation; a third slot carries the reference object on Distance). The interface receives their normalized coordinates, read from the backbone’s own attention (Sec. 3), and nothing else. Each is lifted to a Fourier feature over six octaves and combined with a learned embedding of the object’s mention slot.

slot 0 being the first-mentioned object, 1 the second, 2 the reference. The module accepts an optional content vector per object, it is disabled in every run reported here, so queries and keys see position and role and nothing else. The bottleneck is therefore enforced twice: the values are content-independent by construction, and the inputs carry no appearance to begin with.

Relational cross-attention. With four heads of width 64 is a bank of learned symbols, sliced to its first N rows and split across heads. Queries and keys come from the same set: the Abstractor’s self-relational case.

The symbols are indexed by slot rather than tied to an object, exchanging c_1 and c_2 changes A instead of permuting it. That is only the capacity for asymmetry and it is used on the counterfactual twins of B.3.

Readout. The Abstractor emits N abstract states into a decoder trained beside it. We need one vector, for one slot, in a decoder that will never be trained. A single learned query $q_r \in \mathbb{R}^{1 \times 256}$ collapses the block

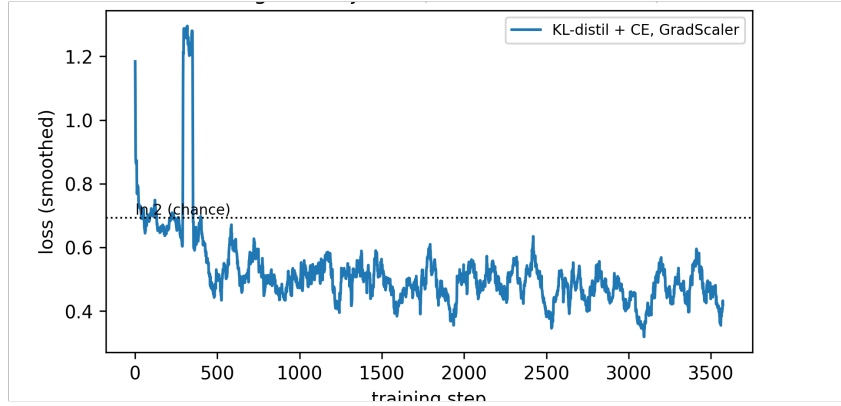


Figure 5: **Training the interface.** Distilling BRIEF_2D, with coordinate-swap augmentation allows our loss curve to converge.(B.3).

The relational computation, the encodings, W_q , W_k , the symbols, the pooling query is 140K parameters. The output MLP that lifts a 256-dimensional relational code into the backbone’s 4096-dimensional hidden space is 17.8M, 99% of the module’s 17.9M.

B.3 Training

Only the interface trains; the backbone is frozen and in eval mode throughout.

Distillation supplies the direction. BRIEF_2D reaches the answer through the same residual stream (Sec. 4.2), which makes its next-token distribution both a dense target and a demonstrably reachable one. We cache the teacher’s logits t and train the interface-injected logits z , taken at the same position, against

$$\mathcal{L} = T^2 \text{KL}(\text{softmax}(t/T) \parallel \text{softmax}(z/T)) + \lambda \text{CE}(z, y), \quad T = 2, \quad \lambda = 0.5.$$

The loss falls from 5.24 to 0.47 and the KL term from 4.22 to 0.05 over four epochs (Fig. 5).

Counterfactual twins. Every two-choice item is paired with a twin whose two coordinates are exchanged and whose label and teacher logits are both taken from the swapped configuration. The pair is separable only by a module that represents the relation asymmetrically, so nothing swap-invariant fits the training set at all. This doubles the training data, and it is what makes the swap control at test time a test rather than a formality.

Hyperparameters. AdamW, learning rate 3×10^{-4} , weight decay 10^{-4} , gradient clipping at 1.0, one item per step, four epochs over the 454 training items (908 with twins, shuffled together). The reported checkpoint is the epoch with the best accuracy on the 65 validation items. Nothing else is selected on validation: the injection layer, β , and the module’s widths are fixed prior.